

Jannatul Ferdous Srabonee

(628) 268 9404 • jfsrabonee@hotmail.com

[Google Scholar](#) | [LinkedIn](#) | [Portfolio](#)

Merced, California, USA

Research Interests: Human-Computer Interaction (HCI), Applied Artificial Intelligence (AI), Accessibility, Brain-Computer Interface (BCI)

EDUCATION

PhD Candidate in Electrical Engineering & Computer Science

Aug 2023 - Present

University of California, Merced (UC Merced)

Merced, CA, USA

Major: Human-Computer Interactions

CGPA: 3.78/4.0

Advisor: Ahmed Arif

Dissertation: Novel Input and Interaction Techniques for Motor and Vision Impaired Individuals

BSc in Computer Science & Engineering

Dec 2015 – Mar 2020

Khulna University of Engineering & Technology (KUET)

Khulna, Bangladesh

Advisor: Muhammad Aminul Haque Akhand

RESEARCH EXPERIENCE

Graduate Student Researcher

Aug 2023 – Present

Inclusive Interaction Lab, UC Merced

Merced, CA, USA

Research Engineer

Feb 2022 – Jul 2023

Advanced Intelligent Multidisciplinary Systems Laboratory

Dhaka, Bangladesh

Machine Learning Engineer Intern

Jul 2021 – Dec 2021

Robotech Valley

Dhaka, Bangladesh

Undergraduate Researcher

Jan 2019 – Mar 2020

Intelligence Systems Lab, KUET

Khulna, Bangladesh

PUBLICATIONS

Conference Papers

[C.3] BezelBraille: A Braille-Inspired Bezel-Guided Virtual Keyboard for Low Vision and Blind Users

Isna Alfi Bustoni, **Jannatul Ferdous Srabonee**, Gulnar Rakhmetulla, & Ahmed Arif

MobileHCI 2026 [[Paper](#)]

[C.2] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription

Jannatul Ferdous Srabonee, Ohoud Mosa Alharbi, Wolfgang Stuerzlinger, & Ahmed Arif

CHI 2025 [[Paper](#)] [[Presentation](#)]

[C.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data

Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique

IEEE TENSYP 2020. [[Paper](#)]

Journal Articles

- [J.2] Computer Vision-Based IoT Architecture for Post COVID-19 Preventive Measures
Ahsanul Akib, Kamruddin Nur, Suman Saha, **Jannatul Ferdous Srabonee**, M. F. Mridha
Journal of Advances in Information Technology (**JAIT**) 2023. [[Article](#)]
- [J.1] Autism Detection from 2D Transformed EEG Signal using Convolutional Neural Network
Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique
Journal of Computer Science (**JCS**) 2022. [[Article](#)]

Book Chapter

- [B.1] Alcoholism Detection from 2D Transformed EEG Signal
Jannatul Ferdous Srabonee, Zahrul Jannat Peya, MAH Akhand, Nazmul Siddique
International Joint Conference on Advances in Computational Intelligence (**IJCACI**), Springer
2020. [[Chapter](#)]

PROJECTS

Eyes-free Text Entry in Smartphones Using a Braille Inspired Virtual Keyboard [Paper](#)

- Improved blind/low-vision users' smartphone text entry speed by 40% and braille proficiency of novices from 12% to 79% across an 8-session study by designing and prototyping a braille-inspired eyes-free keyboard
- Conducted a user study with 12 blind and low-vision participants
- To appear at MobileHCI 2026

Depression Screening Tool Using BCI [Code](#)

- Recorded and preprocessed brain-signal data (EEG) using Brain Products' 64-channel EEG device and Brain Vision software
- Built a brain-signal-based (EEG) depression screening framework, achieving 99% classification accuracy by applying data science and fine-tuning machine learning model (SVM)
Earned national recognition by receiving the Winner Award out of 292 finalist projects at the BASIS National ICT Awards, Bangladesh

Face Mask Detection to Address COVID-19 Mask-Compliance Gaps [Paper](#) [Code](#)

- Built a low-cost IoT prevention system combining face mask detection, contactless temperature monitoring, and automated disinfectant dispensing, using deep learning (MobileNetV2)
- Achieved high accuracy of 99.88% in detecting masked, improperly masked, and unmasked faces in real-time video streams
- Published in Scopus-indexed journal

Early Autism Detection from Brain-Signals Using Deep Learning [Paper](#) [Code](#)

- Improved brain-signal-based (EEG) autism detection by applying correlation-based transformation of brain-signal (EEG) and deep learning (CNN)
- Achieved 88% classification accuracy using five-fold cross-validation
- Published in Scopus-indexed journal

EEG-based Alcoholism Detection [Paper](#)

- Developed an EEG-based alcoholism detection system using deep learning model (CNN)
- Achieved 98.13% while tested on a benchmark EEG dataset
- Published in Springer

POSTERS & PRESENTATIONS

Posters

[Po.2] AI in Accessibility, Research Week, University of California, Merced. **2026.**

[Po.1] EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis, EBRAINS Workshop: Brain Activity across Scales and Species: analysis of Experiments and Simulations, Rome, Italy. **2022.** (Virtual)

Presentations

[Pr.8] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription, Yokohama, Japan, **CHI 2025.**

[Pr.7] A Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians, Bangladesh Association of Software Information Services National ICT Awards, Dhaka, Bangladesh, **BASIS 2022.**

[Pr.6] Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians, International Conference on 4th Industrial Revolution for the Emerging Future, Dhaka, Bangladesh, **4IR 2022.**

[Pr.5] EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis, EBRAINS Workshop: Brain Activity across Scales and Species: analysis of Experiments and Simulations, Rome, Italy, **2022.** (Virtual)

[Pr.4] COVID-19 Defense System, **IEEE YESIST12** Maker Fair, Dhaka, Bangladesh, **2021.**

[Pr.3] Mobile App for Post COVID-19 Measures, NASA Space Apps Challenge, Dhaka, Bangladesh, **2021.**

[Pr.2] Alcoholism Detection from 2D Transformed EEG Signal, International Joint Conference on Advances in Computational Intelligence, Daffodil International University, Dhaka, Bangladesh, **IJCACI 2020.**

[Pr.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data, Dhaka, Bangladesh, **IEEE TENSYP2020.**

TEACHING EXPERIENCE

Teaching Assistant, UC Merced

Aug 2023 – May 2026

CSE 100: Algorithm Design and Analysis

CSE 155: Introduction to Human-Computer Interaction

CSE 022: Introduction to Programming

CSE 015: Discrete Mathematics

CORE SKILLS

Languages: Python, C/C++, MATLAB, JavaScript, HTML, CSS

AI/ML Frameworks: TensorFlow, Keras, scikit-learn, OpenCV

Data Manipulation & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Research Methods: User Studies, Usability Testing, Experimental Design, Mixed-Methods Research, Interviews, Focus Groups, Quantitative Data Analysis, Statistical Hypothesis Testing

FELLOWSHIPS & AWARD

- EECS Bobcat Fellowship | \$11,000, (2024, 2025, 2026), University of California, Merced, CA, USA
- Winner Award, 2022, Bangladesh Association of Software Information Services (BASIS) National ICT Awards, Dhaka, Bangladesh
- General Scholarship in Higher Secondary School Certificate examination, 2015, Board of Intermediate and Secondary Education, Dhaka, Bangladesh

ACADEMIC SERVICE

Co-mentor of Undergraduate Interns Inclusive Interaction Lab at UC Merced Emeka Sha (CSE), Anvita Kisara (CSE, graduated)	Nov 2024 – Present <i>Merced, CA, USA</i>
Reviewer at CHI Articles Reviewed: 1	Jan 2026 – Present
Student Volunteer CHI 2025	Apr 2025 - May 2025 <i>Yokohama, Japan</i>
Senior Executive Committee Member IEEE Student Branch KUET	Feb 2019 - Feb 2020 <i>Khulna, Bangladesh</i>

PROFESSIONAL AFFILIATIONS

Association for Computing Machinery (ACM)	since Sep 2023
Health Sciences and Research Institute (HSRI) at UC Merced	since Nov 2024
***Completed ASSETS Future Program Committee workshop	Mar 2026

PROFESSIONAL EXPERIENCE

Python Developer Mir Info Systems Ltd	Jun 2022 – Oct 2022 <i>Dhaka, Bangladesh</i>
---	--